EE655 Course Project

Novel Approaches Tackling Sketch To Image Generation

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Abstract

Sketch-to-image synthesis is a challenging problem that requires bridging sparse structural input with rich semantic image generation. In this work, we explore and compare three paradigms-CycleGAN, CLIP-guided Stable Diffusion, and conditional GANs-for translating sketches into realistic images. We propose a novel hybrid pipeline that leverages CLIP to automatically extract semantic prompts from sketches and combines them with ControlNet-based diffusion for structure-preserving image synthesis. Unlike conventional approaches that rely solely on paired datasets or manual prompting, our method introduces an automated sketch-to-prompt mechanism along with dual conditioning using both textual and structural cues. Additionally, we introduce a data bootstrapping strategy that generates synthetic sketch-image pairs using edge detection and inverse GAN mappings, enabling training in domains with limited paired data. Our experiments show that diffusion-based methods provide superior generalization and visual fidelity, while GAN-based approaches offer finer control in domain-specific settings. This highlights the trade-off between flexibility and controllability in sketch-to-image generation.

1 Introduction

Sketch to image is an important problem to tackle since it has many applications spanning over many domains - creative, industrial, educational and technical.

1. **Bridging the Gap Between Imagination and Realism:** Sketches are fast, intuitive, and universal ways of communicating ideas. Generating images from sketches allows people to visualize ideas without artistic skill, quickly iterate on concepts, and communicate visually with machines.
2. **Empowering Non-Experts in Design and Art:** Sketch-to-image systems enable users such as product designers, students, or everyday users to create realistic visuals without the need for professional tools, democratizing creative expression. Further it helps develop the base for more complicated operations based on sketches such as 3D Model generation.
3. **Insights on Historical Sketches:** This technology can be leveraged to possibly provide insights and information in a better manner on sketches with historical relevance, and help with visual reconstruction, research etc.
4. **Challenging and Rich AI Problem:** The task involves abstract and sparse input, one-to-many mappings, semantic understanding, and fine-grained control-making it a valuable benchmark for AI research in generative modeling and multimodal understanding.
5. **Practical Applications:** This technology can be applied in entertainment (e.g., game design), fashion (e.g., clothing try-ons), architecture, e-commerce, and even forensics (e.g., generating faces from police sketches).
6. **Advancing Human-AI Collaboration:** Sketch-to-image tools facilitate co-creation, allowing humans to provide creative intent while AI helps with rendering, refinement, and exploration of variations.
7. **Cross-Cultural, Language-Agnostic Potential:** Since sketching is a universal form of expression, these tools can be used across different languages and cultures, making interfaces more intuitive and globally accessible.
8. **Dataset Scarcity Pushes Innovation:** The lack of large paired datasets encourages the development of innovative techniques such as few-shot learning, domain adaptation, and cross-

modal generalization.

Novelty: In the two primary approaches we have used, the novelty is demonstrated in both approaches in a different sense. In the CLIP-guided Stable Diffusion, we try to leverage the power of having pre-trained models with very, very large context to serve a specific purpose using fine-tuning and guided diffusion to generate images retaining similarity to image sketch, despite using an unrelated algorithm. On the other hand, the method leveraging GANs faces various challenges - it requires training a model from scratch since fine-tuning a GAN serving a different purpose is unlikely to yield good results, training them can be very resource intensive and it also requires a large dataset since we are doing operations going from image-to-image.

2 Proposed Architecture

2.1 CLIP-Guided Conditioned Stable Diffusion

In this approach, we tried to target the problem of avoiding training large models due to resource limitations, and rather provide a flexible, out-of-the-box method for sketch to image generation, leveraging pre-trained models, with fine-tuning applied over them, to give a solution which can synthesise images from almost any domain without need for retraining.

2.1.1 Step 1: CLIP based Sketch to Prompt generation

In the first step of our pipeline, we leverage the Contrastive Language-Image Pre-training (CLIP) model to extract semantic information from the given sketch and convert it into a descriptive text prompt. CLIP, trained to understand both images and text in a shared embedding space, can compute the similarity between a given image and textual descriptions. In this context, we apply CLIP to the sketch to generate a high-level prompt that describes the visual content of the image.

Specifically:

Input: A sketch or line drawing representing a scene or object.

CLIP Processing: The sketch is passed through CLIP's visual encoder, which embeds the image into a high-dimensional vector space. Simultaneously, a set of candidate textual prompts is evaluated through CLIP's textual encoder to identify the closest matching description.

Output: The most semantically relevant text prompt is selected based on the cosine similarity between the sketch's embedding and the textual candidates, forming an initial prompt that summarizes the sketch's content.

2.1.2 Step 2: Prompt Refinement and Enhancement

In the second step of the pipeline, the initial text prompt generated by CLIP undergoes refinement and enhancement based on the style selected. The purpose of this step is to enhance the quality and relevance of the prompt by focusing on scene coherence and descriptive clarity, rather than just reflecting the sketch's features.

Specifically:

Input: The initial prompt generated by CLIP, which may still be vague or overly focused on the sketch's technical attributes (e.g., "a pencil drawing of a tree").

Processing: We take the most relevant features of the CLIP interrogator prompt and add other enhancements to the prompt which makes the prompt more vivid and helps the ControlNet model to generate the image more sketch-like while also adding a creative touch.

We add the chosen style-related features, Quality boosters and also Composition enhancers (like depth of field, lightings, and shadows.

It removes any references to the sketch itself (such as "drawing," "sketch," or "lines") and shifts the focus to the actual content, ensuring that the prompt describes a more realistic or idealized version of the scene.

Output: A refined prompt that describes the scene with rich detail, such as textures, lighting, and spatial relationships, while aligning with the original conceptual intent of the sketch.

2.1.3 Step 3: Image Generation Using ControlNet and Text-to-Image Diffusion Model

In the third step of our pipeline, the refined textual prompt from the LLM is used to guide the image generation process. Here, we leverage the ControlNet model in combination with a text-to-image diffusion model (e.g., Stable Diffusion) to synthesize a high-quality image that matches both the original sketch and the idealized scene described by the refined prompt.

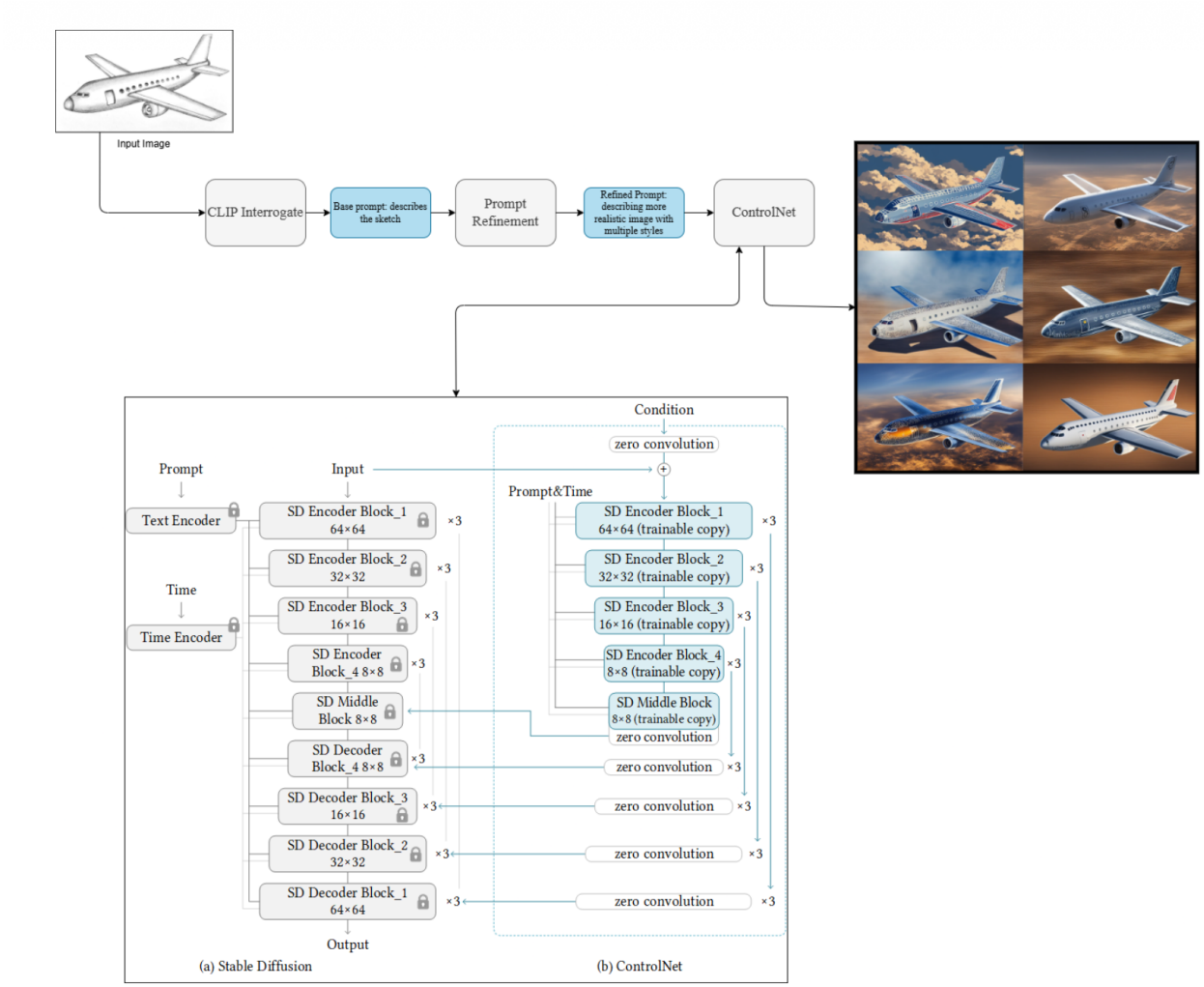

Figure 1: Proposed Architecture and Pipeline for the CLIP-Guided Conditioned Stable Diffusion

Specifically:

Input:

- The refined textual prompt generated in Step 2, which describes the scene in rich, vivid detail.
- The original sketch or its processed features, which act as a control signal for conditioning the generated image.

ControlNet Processing:

ControlNet is a model that enables the conditioning of the diffusion model based on additional input signals, such as edge maps, depth maps, or other structured visual information. We leveraged the pretrained models for Canny Edge, Scribble and Soft-Noise for our pipeline.

In this step, ControlNet takes the original sketch as a control input and uses it to guide the image generation process, ensuring that the generated image retains the spatial layout, proportions, and key structural elements from the sketch.

The role of ControlNet is to allow the diffusion model to follow specific geometric cues (e.g., edges, contours, or shapes) while generating the image, ensuring that it remains faithful to the initial sketch's composition.

Text-to-Image Diffusion Model (e.g., Stable Diffusion):

The text-to-image model (such as Stable Diffusion) is a powerful generative model trained to synthesize realistic images from textual descriptions.

The refined prompt, now rich in visual details, is used as the primary conditioning input for the diffusion

model, guiding the generation of photorealistic textures, lighting, and scene features.

The control signal (provided by ControlNet) helps ensure that the model respects the structural layout of the sketch, aligning the generated image's content with both the initial visual structure and the idealized textual description.

Output: A fully synthesized image that blends the structural elements from the sketch with the enhanced visual details described by the LLM. The final image should reflect a realistic, detailed interpretation of the scene as per the refined prompt, while maintaining the essential composition and features from the original sketch.

For example, if the refined prompt is "a wooden treehouse nestled in the branches of a tall oak tree, surrounded by green foliage and bathed in soft afternoon light," the diffusion model will generate an image of a realistic treehouse scene, with intricate textures, lighting, and natural elements, all while keeping the layout and positioning of the treehouse from the original sketch intact.

2.2 GAN Based Approach:

One of the largest challenges in this approach is the requirement of a large dataset for training GANs. We planned to tackle this by introducing novelty in obtaining the way we obtained the dataset for training and our approach for training.

- **Dataset Generation** A lot of domains do not have datasets of (image, sketch) pairs. Therefore, we felt it was important to devise a method to generate datasets in domains where such data might not be available. To tackle this, we used generic datasets of images of a domain (for example, human faces) - and generated image, sketch pairs using a few approaches - Canny Edge Detection + Thresholding, Photoshop Filters, etc. These proved to be a great method to train since these retained a lot of features typically present in sketches - edge details, outlines, etc..
- **Data Augmentation** To further generate data for refining the model, we used our GAN trained for the inverse task, to help introduce more data. The approach is explained ahead in our CycleGAN based approach.

We tested out two GAN based models for image-to-image translation - CycleGAN and our own written GAN, similar to pix2pix.

2.3 Sketch-to-Image Translation Using CycleGAN

CycleGAN (Cycle-Consistent Generative Adversial Network) architecture is designed for unpaired image-to-image translation, meaning it learns to translate images from one domain to the another without requiring paired examples. Although we were successful in establishing a paired image database, we chose to try CycleGAN as well to have a model trained in 2 ways, sketch-to-image and image-to-sketch, and to try and use this for semi-supervised **data augmentation** for generating datasets for further training. We believed that image-to-sketch mapping is a less complex task than the inverse. Images lose colour data, detail, depth etc. in the inverse, and hence we hypothesized that image-to-sketch generator within CycleGAN could be trained effectively with a relatively smaller dataset. We then used the synthetic sketch-image pairs for **augment-training** of the sketch-to-image model.

2.4 Sketch-to-Image Translation Using custom GAN

Objective

This implements an sketch-to-image translation pipeline using a Generative Adversarial Network (GAN) based on the Pix2Pix architecture (the backbone is similar) but we are using custom upsample and downsample blocks. The primary goal is to train a model that learns a mapping from input images to corresponding target images, useful for tasks such as image colorization, sketch-to-photo generation, or domain conversion.

Generator (U-Net based architecture) The generator is designed as a U-Net architecture: Uses downsample and upsample functions with Conv2D and Conv2DTranspose layers. Includes skip connections to preserve spatial detail. Activation: Final tanh activation scales output to $[-1, 1]$.

Discriminator Takes both the input and target/generated image. Outputs a matrix of probabilities to assess real vs fake image patches. The discriminator uses binary cross entropy loss.

Generator Loss Combination of:

Adversarial Loss: Encourages the generator to fool the discriminator.

L1 loss: penalizes pixel-wise difference between generated and target images.

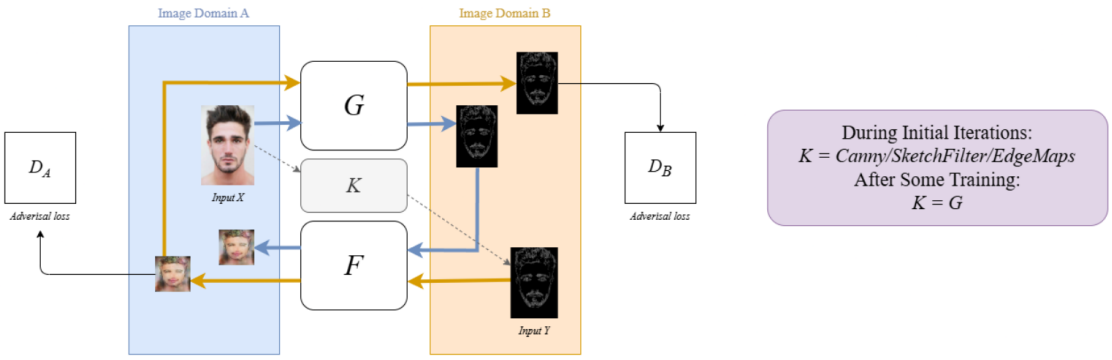

Figure 2. Proposed Architecture and Pipeline for the CycleGAN based Approach

Figure 2: Proposed Architecture and Pipeline for the CycleGAN based Approach

Figure 3: Proposed Architecture and Pipeline for our custom GAN based Approach

Training Procedure

- Batch Size: 4
- Image size: 128×128
- Epochs: 10

increased batch size for parallel computing and reduce training time. there are about 7300 images in training dataset so about 1800 steps in each Epoch.

then we compute losses and update both the generator and discriminator using gradient descent. Training Loop: In each train step we:

1. Generate fake output using the generator.
2. Pass real and fake pairs to the discriminator.
3. Compute and apply gradients to update both models.

Optimizers Both models use the Adam optimizer here we are using learning rate of 2×10^{-4}

The biggest challenge we faced with the whole GAN method was the lack of hardware resources to train and test the models promptly. It took the order of 15+ hours for each iteration of training on a 7.2k image dataset with around 10 epochs, due to very constrained hardware availability. We fell short in being able to experiment a lot due to this restriction.

3 Results and Conclusion:

Table 1 shows a comparison of the various key criteria relating to the methods we have employed. We find

that ControlNet is a more general model and can be used more generally due to the ease of availability of pre-trained models, however, it provides lesser fine-grained control and requires additional prompting to help style it better and give more context. It can also go wrong with recognition of classes and styling. GAN based approaches, are not as readily available, and fine-tuning is not a feasible option unless relevant models can be found. Training these takes time and large amounts of data and hence seems less feasible. However, with further training, we believe GANs will provide more fine-grained control over image generation with context specific to domains, and could potentially outperform in some areas. They also require lesser resources for running, since they target specific image-to-image translations. However, they will always lack customizability, ease of editing, etc. without changing the architecture.

3.1 Some examples using ControlNet:

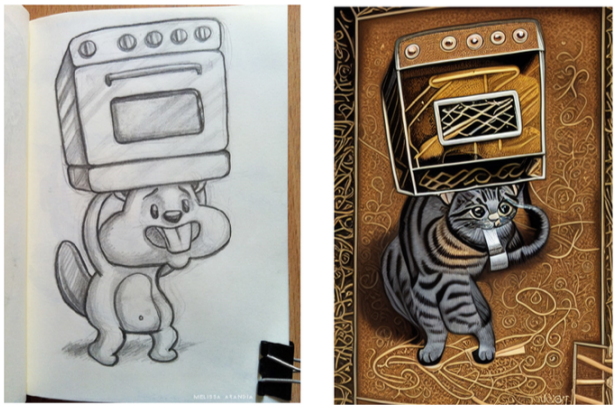

Figure 4: CLIP-Guided Conditioned Stable Diffusion

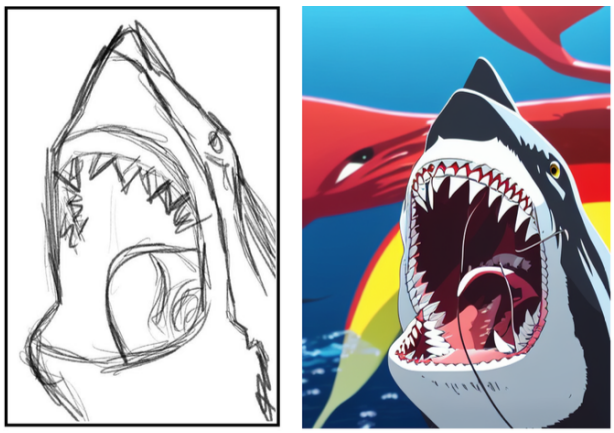

Figure 5: CLIP-Guided Conditioned Stable Diffusion

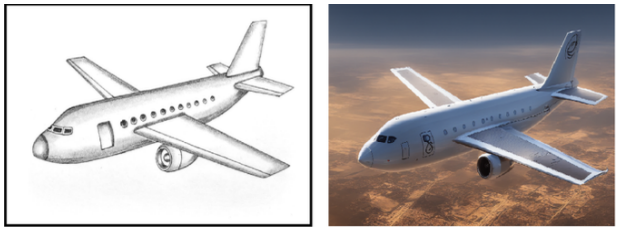

Figure 6: CLIP-Guided Conditioned Stable Diffusion

3.2 Some results using GAN based approach

Figure 7: Custom GAN based input output

Figure 8: Custom GAN based input output

Figure 9: Custom GAN based input output

3.3 Our Code and Datasets Used:

3.3.1 CodeBase

CodeBase - This is our codebase.

3.3.2 Datasets

Human Face - This is our main data set on which we trained our GAN model.

Sketchy Database - GATech - Another large collection of basic sketches and corresponding images, which we trained partially on.

References

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Table 1: Comparison of ControlNet-based and GAN-based Sketch-to-Image Models

Criteria	**ControlNet-based Model**	**GAN-based Model**
Architecture Base	Built on diffusion models (e.g., Stable Diffusion) using additional control branches	Uses adversarial training with generator and discriminator
Input Conditioning	Supports multiple inputs (sketch, depth, canny, pose, etc.) via external encoders	Typically limited to sketch and optional labels or style vectors
Training Stability	Generally stable due to denoising process of diffusion	Less stable; prone to mode collapse
Output Quality	High-fidelity and detailed; benefits from pretrained backbones	Sharp outputs but may lack consistency depending on architecture
Style Control	Easily guided by text prompts, latent fusion, or ControlNet fusion	Style control requires embeddings or explicit architecture changes
User Interactivity	Supports text+sketch fusion, interactive editing (e.g., inpainting, guidance overlays)	Non-interactive; limited editing without retraining
Inference Time	Slower due to iterative denoising	Fast, single forward pass
Flexibility & Modularity	Highly modular—easy to add or switch control branches	Less modular—adaptation often needs new training
Data Efficiency	Can fine-tune on small datasets using large pretrained models	Needs large sketch-image datasets to generalize well
Typical Use Cases	Creative image synthesis, concept art, controllable generation	Style transfer, domain mapping, real-time sketch translation

Individual Contributions

Approach 1: GAN Architecture and Data Synthesis (Sarvesh Bharambe & Harshvardhan Agarwal)

- Addressed the fundamental challenge of dataset scarcity by generating synthetic image-sketch pairs using generic datasets and techniques such as Canny Edge Detection and Photoshop filters.
- Implemented the CycleGAN architecture for unpaired image-to-image translation to facilitate semi-supervised data augmentation.
- Designed the custom GAN model based on the Pix2Pix architecture, configuring the U-Net generator, custom upsample/downsample blocks, and the discriminator.
- Managed the computationally intensive training procedures, optimizing adversarial and L1 loss functions to navigate severe hardware limitations and extensive training times.

Approach 2: CLIP-Guided Conditioned Stable Diffusion (Vrushabh D Undri& Daksh Pratap Singh)

- Implemented the Contrastive Language-Image Pre-training (CLIP) model to extract semantic information from sketches and convert them into descriptive text prompts.
- Engineered the prompt refinement pipeline to enhance scene coherence by adding composition enhancers and removing literal sketch references.
- Integrated the refined textual prompts with a text-to-image diffusion model to guide the synthesis of high-fidelity, photorealistic images.
- Configured ControlNet using pre-trained models (Canny Edge, Scribble, Soft-Noise) to ensure the generated outputs strictly respected the spatial layout and structural elements of the original sketches.

Shared Collaborative Efforts (All Team Members)

- Authored the comprehensive project abstract, problem introduction, and literature compilation.
- Conducted rigorous cross-model evaluations assessing the strengths and limitations of each method in terms of fidelity, diversity, and semantic alignment.
- Synthesized experimental findings to compile the comparative analysis matrix and formulate the final project observations and conclusions.